

RITI SINGH

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EDUCATION

Dartmouth College

Expected June 2027

Bachelor of Electrical and Computer Engineering

GPA: 3.70/4.00

Related Coursework: Software Design and Implementation, Embedded Systems, Digital Electronics, Algorithms/Data Structures

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Bash, MATLAB

Software & Infrastructure: FastAPI, REST APIs, Linux, Git, Docker, AWS, CI/CD

Automation & Testing: Test Automation, System Validation, Data Pipelines, Failure Analysis, Regression Testing

Systems & Hardware: TCP/IP, Wireless Systems, STM32, FPGA/VHDL, Sensors & Instrumentation, Hardware-Software Integration

Data: Neo4j, Amazon Neptune, Pandas

WORK EXPERIENCE

EchoStar | Boost Mobile Wireless Device Engineering Team

Denver, CO

Platform (AI/Automation) Engineering Intern

Summer 2026

- Partnered with wireless engineers and SMEs to identify high-friction testing and validation workflows, then built a Python-based engineering platform that reduced workflow time and manual effort by 95%.
- Developed backend services, REST APIs, and automated data pipelines using Python, FastAPI, AWS, Neptune, and Neo4j, integrating 12,000+ specification nodes, 606 requirements, and 728 test cases.
- Built automated regression and system-validation workflows to test end-to-end functionality, diagnose failures, and improve reliability across wireless engineering processes.
- Diagnosed system and retrieval failures through edge-case testing and user feedback; presented the platform to engineering leadership and earned 2nd Place in EchoStar's company-wide competition.

Smart Sort

Hanover, NH

Electrical and Circuits Lead, Business Lead

March 2024 – September 2025

- Co-led development of an autonomous waste-sorting system, owning electrical architecture and hardware-software integration from prototype through functional system testing.
- Integrated inductive and weight sensors, microcontroller control logic, electromechanical actuators, relays, and power electronics, debugging system-level failures across sensing, control, and actuation.
- Received the Philip R. Jackson Award for Innovation.

EvergreenAI

Hanover, NH

Project Assistant - AI Engineering

September 2025 – June 2026

- Developed structured validation and edge-case testing workflows for production AI systems, analyzing failure modes and building repeatable evaluation datasets to improve system reliability.

Dartmouth College | EE Just Fellowship

Hanover, NH

EE Just Research Fellow

Summer 2024

- Built Python data pipelines processing 10,000+ observations across multiple datasets and performed statistical analysis of recommendation-system behavior.

SELECTED PROJECTS

Relay - Agentic Network Incident Response - • Built a stateful network-operations platform that investigates connectivity failures using topology, telemetry, logs, and diagnostic tools, identifies root causes, and proposes human-approved remediation.
• Developed deterministic infrastructure failure scenarios and automated verification/evaluation workflows for routing, interface, DNS, ACL, and configuration failures.

Spectrace — Engineering Traceability & Graph Analysis Platform - Built a full-stack engineering platform with typed Python backend services, REST APIs, graph-based data modeling, and containerized infrastructure for analyzing relationships across technical requirements and test artifacts.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP EXPERIENCE

Leadership: CORE Risk Manager, Kappa Delta Epsilon | President & Outreach/Support Team Lead, Sexual Assault Peer Alliance | Peer Counselor, Teen Lifeline (1,000+ youth supported)

Awards: 2nd Place, EchoStar AI Innovation Competition (2026) | Philip R. Jackson Award for Innovation | Dartmouth College Honors List | Congressional Leadership Award | NSDA Academic All-American | Junior Achievement Arizona "18 Under 18"